
Week 35: Discrete-time Markov chains

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These notes are written by me, so errors may and will occur. When in doubt, trust the book and Gunnar's lectures!

1. Motivation

See [Veritasium: The Strange Math That Predicts \(Almost\) Anything](#) for a nice video on the topic.

Some research questions we might attempt to answer:

- How much uranium do we need to build a nuclear bomb?
- How many people will get Covid-19 within the next 6 months?
- How many times do we need to shuffle a deck of cards before it is sufficiently random?

Some interesting applications of Markov chains:

- **Google's PageRank algorithm:** Google's original PageRank algorithm models a person randomly clicking from one webpage to another. The long-run probability of being on each page provides a measure of how important that page is.
- **Diffusion models for generative AI:** Image generators based on diffusion gradually add noise to an image and then learn how to reverse this process to generate new images. We can interpret the state of the image as a Markov chain.
- **Language models and early text generation:** Markov models were commonly used to generate text by predicting the next word. However, LLMs are likely a better choice nowadays.
- **Monte Carlo and MCMC methods:** Very important in physics and machine learning, Markov chains are used to sample from complex distributions.
- **Hidden Markov models:** Used in many applications, for example speech recognition, robotics, and GPS systems.
- **Gaussian Markov Random Fields:** A computationally efficient model that I worked on during my [master's thesis](#).

2. Theory

We will go through the basics of discrete-time Markov chains and perform first-step analysis. Next week we will explore the long-term behavior of Markov chains.

2.1. Definitions and basic concepts

Definition 2.1. (*Discrete-time stochastic process*) A discrete-time stochastic process is a family of random variables $\{X_t : t \in T\}$ where T is discrete.

We call X_t the *state* at time t , and the set of all possible states the *state space*. A typical choice of index set is $T = \{0, 1, \dots\}$.

Definition 2.2. (*Discrete-time Markov chain*) A discrete-time Markov chain is a discrete-time stochastic process $\{X_t : t \in 0, 1, \dots\}$ that satisfies the *Markov property*

$$P(X_{t+1} = j \mid X_0 = i_0, \dots, X_t = i_t) = P(X_{t+1} = j \mid X_t = i_t),$$

for $t = 0, 1, \dots$ and all states $i_0, \dots, i_t, j$ for which the conditioning event has positive probability.

Example 2.3. (*Markov chains for finite and infinite state spaces*)

- Consider a student in Trondheim who is either at home (H), at Gløshaugen (G), or at Samfundet (S). The student moves between these three locations, so the state space is $\{H, G, S\}$.
- Consider the number of unread messages a student has at the end of each day. New messages may arrive and old messages may be read, so the number changes from day to day. Let X_t denote the number of unread messages on day t . The state space is $\{0, 1, 2, \dots\}$.

Definition 2.4. (*One-step transition probabilities*) For a discrete-time Markov chain $\{X_t : t \in 0, 1, \dots\}$, we call

$$P_{i,j}^{t,t+1} = P(X_{t+1} = j \mid X_t = i)$$

the *one-step transition probabilities* from state i to state j at time t .

From now on, we always assume *time-homogeneous* transition probabilities, meaning $P_{i,j}^{t,t+1} = P_{i,j}$ for all $t = 0, 1, \dots$ and all states i and j .

Example 2.5. The Markov property and time-homogeneity simplify transition probabilities give some nice simplifications. See that for all $t = 0, 1, \dots$

- $P(X_{t+1} = j \mid X_t = i) = P_{i,j}^{t,t+1} = P_{i,j} = P_{i,j}^{0,1} = P(X_1 = j \mid X_0 = i)$.
- $P(X_{t+1} = j \mid X_0 = i_0, \dots, X_t = i) = P(X_{t+1} = j \mid X_t = i) = P_{i,j}$.
- By repeatedly applying Bayes' theorem, we also find

$$\begin{aligned} P(X_0 = i_0, \dots, X_t = i_t) &= P(X_0 = i_0)P(X_1 = i_1 \mid X_0 = i_0) \cdot \dots \cdot P(X_t = i_t \mid X_{t-1} = i_{t-1}) \\ &= P(X_0 = i_0)P_{i_0,i_1} \cdot \dots \cdot P_{i_{t-1},i_t}. \end{aligned}$$

Definition 2.6. (*Transition diagram*) Let $\{X_t : t = 0, 1, \dots\}$ be a discrete-time Markov chain. A *state transition diagram* visualizes the transition probabilities as a weighted directed graph, where the nodes are the states and the edges are the possible transitions marked with the transition probabilities.

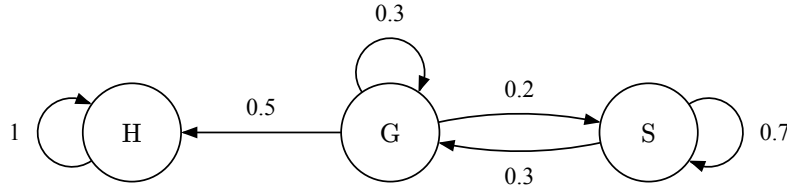
Definition 2.7. (*Transition probability matrix*) For a discrete-time Markov chain with a finite state space $\{0, 1, \dots, N\}$, we call

$$\mathbf{P} = \begin{bmatrix} P_{0,0} & P_{0,1} & \dots & P_{0,N} \\ P_{1,0} & P_{1,1} & & \vdots \\ \vdots & & \ddots & \\ P_{N,0} & \dots & & P_{N,N} \end{bmatrix}$$

the *transition probability matrix*.

For an infinite state space $\{0, 1, \dots\}$, we can envision an infinitely-sized matrix. Also note that I use the convention of writing matrices in bold $\mathbf{P}$, and the entries of the matrix in normal font $P_{i,j}$.

Example 2.8. If we consider the Trondheim student from [example 3](#), the transition diagram is



and the corresponding transition probability matrix is

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 0.3 & 0.2 \\ 0 & 0.3 & 0.7 \end{bmatrix}.$$

Theorem 2.9. (*n-step transition probabilities*) For a time-homogeneous Markov chain $\{X_t : t = 0, 1, \dots\}$ with countable state space S and any $m \geq 0$, we have

$$P(X_{m+n} = j \mid X_m = i) = P_{i,j}^{(n)} = \sum_{k \in S} P_{i,k} P_{k,j}^{(n-1)}, \quad n > 0, \quad (1)$$

where we define

$$P_{i,j}^{(0)} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

Theorem 2.10. Recognizing that (1) is a matrix multiplication, we can write

$$\mathbf{P}^{(n)} = \mathbf{P} \cdot \mathbf{P}^{(n-1)}, \quad n > 0,$$

and iteratively applying this gives

$$\mathbf{P}^{(n)} = \underbrace{\mathbf{P} \cdot \mathbf{P} \cdot \dots \cdot \mathbf{P}}_n = \mathbf{P}^n.$$

2.2. First-step analysis

First-step analysis is a technique for analyzing Markov chains by breaking down the probabilities that can arise at the end of the first step.

Example 2.11. (*Absorbing state*) Going back to the Trondheim student from [example 8](#), we can see that once the student is at home, they will stay there forever. This means that state H is an *absorbing* state.

Definition 2.12. (*Hitting time*) Let $\{X_t : t = 0, 1, \dots\}$ be a Markov chain, and let A be a set of states. The *hitting time* of A is the random variable

$$T_A = \inf\{t \geq 0 : X_t \in A\}.$$

We specify $\inf \emptyset = \infty$, meaning $T_A = \infty$ if the chain never reaches A .

Definition 2.13. (*Absorbing state*) For a Markov chain, a state i such that $P_{i,j} = 0$ for all $j \neq i$ is called absorbing. A set of states A is absorbing if $P_{i,j} = 0$ for all $i \in A$ and $j \notin A$.

Theorem 2.14. Let $\{X_t : t = 0, 1, \dots\}$ be a Markov chain with state space $S = \{0, 1, \dots, N\}$ and transition probability matrix $\mathbf{P}$. Let $A \subseteq S$ be an absorbing set of states.

- If u_i is the probability of absorption in $j \in A$ starting from $X_0 = i$, then

$$u_i = P(X_{T_A} = j \mid X_0 = i) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \in A \text{ and } i \neq j \\ P_{i,j} + \sum_{k \in A^c} P_{i,k} u_k & \text{otherwise.} \end{cases}$$

- If $v_i = \mathbb{E}[T_A \mid X_0 = i]$ is the expected time until absorption starting from $X_0 = i$, then

$$v_i = \begin{cases} 0 & \text{if } i \in A \\ 1 + \sum_{k \in A^c} P_{i,k} v_k & \text{otherwise} \end{cases} \quad (2)$$

Problem 2.1. (Problem 1 in Exercise 2: Part 1)

Consider the Markov chain $\{X_t : t = 0, 1, 2, \dots\}$ with state space $\{A, B, C\}$ and transition probability matrix given by

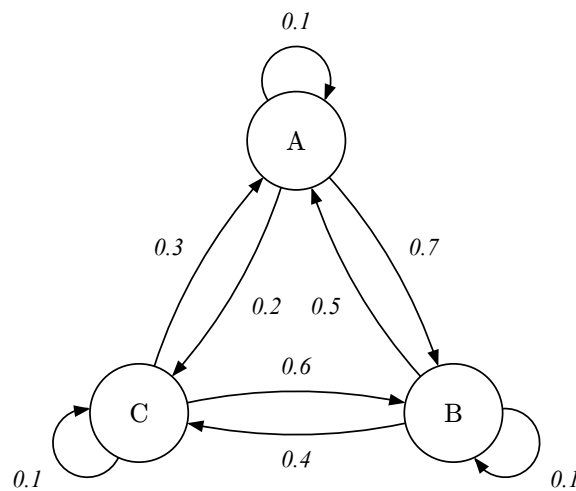
$$\mathbf{P} = \begin{bmatrix} 0.1 & 0.7 & 0.2 \\ 0.5 & 0.1 & 0.4 \\ 0.3 & 0.6 & 0.1 \end{bmatrix}.$$

The probability distribution of the initial state X_0 is given by $P(X_0 = A) = 0.2$, $P(X_0 = B) = 0.5$ and $P(X_0 = C) = 0.3$. Do the following:

- Draw the corresponding transition diagram.
- Compute $P(X_0 = A, X_1 = B, X_2 = C)$
- Compute $P(X_1 = A)$
- Compute $P(X_1 = C)$ (here it is useful to know that $P(X_1 = B) = 0.37$)

Solution.

- The corresponding transition diagram is



- Using Bayes' rule

$$\begin{aligned}
 P(X_0 = A, X_1 = B, X_2 = C) &= P(X_2 = C \mid X_1 = B, X_0 = A) \cdot P(X_0 = A, X_1 = B) \\
 &= P(X_2 = C \mid X_1 = B) \cdot P(X_1 = B \mid X_0 = A) \cdot P(X_0 = A) \\
 &= P_{B,C} \cdot P_{A,B} \cdot P(X_0 = A) \\
 &= 0.4 \cdot 0.7 \cdot 0.2 = 0.056
 \end{aligned}$$

- Using the law of total probability

$$\begin{aligned}
 P(X_1 = A) &= \sum_{k \in \{A, B, C\}} P(X_1 = A \mid X_0 = k) P(X_0 = k) \\
 &= 0.1 \cdot 0.2 + 0.5 \cdot 0.5 + 0.3 \cdot 0.3 \\
 &= 0.36.
 \end{aligned}$$

- $P(X_1 = C) = 1 - P(X_1 = A) - P(X_1 = B) = 1 - 0.36 - 0.37 = 0.27$.

Problem 2.2. (Problem 1 in Exercise 2: Part 2) Consider the same Markov chain as in Problem 2.1. We know that

$$\mathbf{P} = \begin{bmatrix} 0.1 & 0.7 & 0.2 \\ 0.5 & 0.1 & 0.4 \\ 0.3 & 0.6 & 0.1 \end{bmatrix}, \quad \mathbf{P}^2 = \begin{bmatrix} 0.42 & 0.26 & 0.32 \\ 0.22 & 0.6 & 0.18 \\ 0.36 & 0.33 & 0.31 \end{bmatrix}, \quad \mathbf{P}^3 = \begin{bmatrix} 0.268 & 0.512 & 0.220 \\ 0.376 & 0.322 & 0.302 \\ 0.294 & 0.471 & 0.235 \end{bmatrix},$$

and that $P(X_0 = A) = 0.2$, $P(X_0 = B) = 0.5$ and $P(X_0 = C) = 0.3$.

Compute the following probabilities:

- $P(X_3 = A)$
- $P(X_6 = A \mid X_3 = C)$
- $P(X_3 = A \mid X_1 = B, X_0 = A)$
- $P(X_3 = C \mid X_6 = A)$

Solution.

- Using the law of total probability

$$\begin{aligned} P(X_3 = A) &= \sum_{k \in \{A, B, C\}} P(X_3 = A \mid X_0 = k) P(X_0 = k) \\ &= P_{A,A}^{(3)} \cdot P(X_0 = A) + P_{B,A}^{(3)} \cdot P(X_0 = B) + P_{C,A}^{(3)} \cdot P(X_0 = C) \\ &= 0.268 \cdot 0.2 + 0.376 \cdot 0.5 + 0.294 \cdot 0.3 \\ &= 0.3298. \end{aligned}$$

- Using time-homogeneity

$$P(X_6 = A \mid X_3 = C) = P(X_3 = A \mid X_0 = C) = P_{C,A}^{(3)} = 0.294.$$

- Using the Markov property and time-homogeneity

$$P(X_3 = A \mid X_1 = B, X_0 = A) = P(X_3 = A \mid X_1 = B) = P(X_2 = A \mid X_0 = B) = P_{B,A}^{(2)} = 0.22.$$

- Using Bayes' rule, the law of total probability, the Markov property and time-homogeneity

$$\begin{aligned} P(X_3 = C \mid X_6 = A) &= \frac{P(X_6 = A, X_3 = C)}{P(X_6 = A)} \\ &= \frac{P(X_6 = A \mid X_3 = C) P(X_3 = C)}{\sum_{k \in \{A, B, C\}} P(X_6 = A \mid X_3 = k) P(X_3 = k)} \\ &= \frac{P(X_3 = A \mid X_0 = C) P(X_3 = C)}{\sum_{k \in \{A, B, C\}} P(X_3 = A \mid X_0 = k) P(X_3 = k)} \\ &= 0.245 \end{aligned}$$

Problem 2.3. (Exam December 2024) A taxi driver provides service in two zones of a city. Fares picked up in zone A will have destinations in zone A with probability 0.6 or in zone B with probability 0.4. Fares picked up in zone B will have destinations in zone A with probability 0.3 or in zone B with probability x .

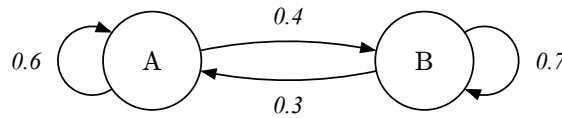
- Explain that $x = 0.7$.
- Write down the transition matrix for the resulting Markov chain and illustrate with a graph with transition probabilities.
- What is the state space? What is the index set of the stochastic process?
- Assume the driver starts in zone A. What is the probability that the driver ends in zone A after two fares? After four fares?

Solution.

- Since the total probability must sum to 1, we have $x = 1 - 0.3 = 0.7$.
- The transition matrix is given by

$$\mathbf{P} = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix},$$

while the corresponding transition diagram is



- As we have two zones, the state space is $\{A, B\}$. The index set of the stochastic process is $\{0, 1, \dots\}$, where X_0 is the driver's initial zone before any fares and X_n is the zone after the n -th fare.
- We have $X_0 = A$, and want to compute $P(X_2 = A \mid X_0 = A)$ and $P(X_4 = A \mid X_0 = A)$. Using the transition matrix, we can compute

$$\mathbf{P}^2 = \mathbf{P} \cdot \mathbf{P} = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix} \cdot \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix} = \begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix},$$

and

$$\mathbf{P}^4 = \mathbf{P}^2 \cdot \mathbf{P}^2 = \begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix} \cdot \begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix} = \begin{bmatrix} 0.4332 & 0.5668 \\ 0.4251 & 0.5749 \end{bmatrix}.$$

Picking the corresponding entries, we find

$$P(X_2 = A \mid X_0 = A) = P_{A,A}^{(2)} = 0.48 \quad \text{and} \quad P(X_4 = A \mid X_0 = A) = P_{A,A}^{(4)} = 0.4332.$$

Problem 2.4. (*Expected time to absorption*) Let $\{X_t : t = 0, 1, \dots\}$ be a Markov chain on the state space $\{0, 1, 2\}$. The transition probability matrix is

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 \\ \alpha & \beta & \gamma \\ 0 & 0 & 1 \end{bmatrix},$$

with $\alpha, \beta, \gamma > 0$ and $\alpha + \beta + \gamma = 1$. Assuming $X_0 = 1$, answer the following

- What is the probability of absorption in state 0?
- What is the expected time until absorption in state 0 or 2? You might need the formula for a geometric series $\sum_{t=0}^{\infty} r^t = 1/(1-r)$.

Solution.

- Letting $A = \{0, 2\}$, we denote $u_i = P(X_{T_A} = 0 \mid X_0 = i)$ for $i \in \{0, 1, 2\}$. Clearly, $u_0 = 1$ and $u_2 = 0$, and the question is how to find u_1 . Using the law of total probability, Bayes' rule and the Markov property, we have

$$\begin{aligned} u_1 &= \sum_{k \in \{0, 1, 2\}} P(X_{T_A} = 0, X_1 = k \mid X_0 = 1) \\ &= \sum_{k \in \{0, 1, 2\}} P(X_{T_A} = 0 \mid X_1 = k, X_0 = 1) P(X_1 = k \mid X_0 = 1) \\ &= \sum_{k \in \{0, 1, 2\}} P(X_{T_A} = 0 \mid X_1 = k) P_{1,k} \\ &= \sum_{k \in \{0, 1, 2\}} u_k P_{1,k} \\ &= u_0 \alpha + u_1 \beta + u_2 \gamma \\ &= \alpha + \beta u_1. \end{aligned}$$

Solving for u_1 gives

$$u_1 = \frac{\alpha}{1 - \beta} = \frac{\alpha}{\alpha + \gamma}.$$

- Let $v_i = \mathbb{E}[T_A \mid X_0 = i]$ for $i \in \{0, 1, 2\}$. Clearly, $v_0 = v_2 = 0$. Starting from state 1, the chain remains in state 1 with probability β , and is absorbed with probability $\alpha + \gamma = 1 - \beta$. Hence, $P(T_A = t) = \beta^{t-1}(1 - \beta)$ for $n = 1, 2, \dots$. Using the formula for the expected value of a discrete random variable, we have

$$\begin{aligned} v_1 &= \sum_{t=1}^{\infty} t P(T_A = t) \\ &= \sum_{t=1}^{\infty} t \beta^{t-1} (1 - \beta) \\ &= (1 - \beta) \sum_{t=1}^{\infty} t \beta^{t-1} \end{aligned}$$

Here we can use a trick by differentiating the geometric series $\sum_{t=0}^{\infty} \beta^t = 1/(1 - \beta)$ to see

$$\sum_{t=0}^{\infty} t \beta^{t-1} = \sum_{t=1}^{\infty} t \beta^{t-1} = \frac{1}{(1 - \beta)^2}.$$

Hence, we have

$$v_1 = \frac{1 - \beta}{(1 - \beta)^2} = \frac{1}{1 - \beta} = \frac{1}{\alpha + \gamma}.$$

3. Towards long-term behavior

One of the powers of Markov chains is that they allow us to study the long-term behavior of a system. For example, considering

$$\mathbf{P} = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix},$$

we have

$$\mathbf{P}^2 = \begin{bmatrix} 0.7 & 0.3 \\ 0.45 & 0.55 \end{bmatrix}, \quad \mathbf{P}^{10} \approx \begin{bmatrix} 0.600391 & 0.3996090 \\ 0.599414 & 0.400586 \end{bmatrix}, \quad \mathbf{P}^{100} \approx \begin{bmatrix} 0.6 & 0.4 \\ 0.6 & 0.4 \end{bmatrix}.$$

Therefore, regardless of the initial state, the probability of being in state 0 after many transitions is approximately 0.6, while the probability of being in state 1 is approximately 0.4. Next week, see study whether the limit

$$\lim_{n \rightarrow \infty} P_{i,j}^{(n)} = \lim_{n \rightarrow \infty} P(X_n = j \mid X_0 = i).$$

exists and, when it does, determine its value.